

Subject Code: AD53XX

Subject Name: MACHINE LEARNING OPERATIONS (MLOps)

Regulation: R25

Year / Semester: II / III

Academic Year: 2026-27

Unit	Topic	PBL Activity & Project Tasks – Implementation / Student Project Task
Unit I	Machine Learning Challenges	PBL Activity & Project Tasks – Implementation: Identify real-world challenges in deploying Machine Learning models such as data drift, model decay, reproducibility, scalability, and monitoring. Analyze how MLOps addresses these challenges. Student Project Task: Select a real-world ML application and document the challenges involved in deploying and maintaining the model in production.
	Machine Learning Operations: Definition, Needs and Challenges	PBL Activity & Project Tasks – Implementation: Understand the fundamentals of MLOps and its role in automating the ML lifecycle, collaboration, and deployment processes. Student Project Task: Prepare a concept map illustrating the MLOps workflow and explain its importance in real-world AI systems.
	Machine Learning vs MLOps	PBL Activity & Project Tasks – Implementation: Compare traditional Machine Learning workflows with MLOps pipelines by analyzing automation, deployment, monitoring, and collaboration practices. Student Project Task: Prepare a comparison table and justify the need for MLOps in production environments.
	DevOps vs MLOps	PBL Activity & Project Tasks – Implementation: Study software DevOps practices and identify additional components required for Machine Learning systems. Student Project Task: Create an architecture diagram highlighting similarities and differences between DevOps and MLOps.
	MLOps Life Cycle	PBL Activity & Project Tasks – Implementation: Design an end-to-end MLOps lifecycle covering data ingestion, preprocessing, model training, deployment, monitoring, and retraining. Student Project Task: Develop a lifecycle flowchart for the selected project using visualization tools.
	Components of MLOps &	PBL Activity & Project Tasks – Implementation: Identify major MLOps components such as data versioning, experiment tracking,

Unit II

MLOps Actors	model registry, CI/CD, deployment, and monitoring along with stakeholder responsibilities. Student Project Task: Assign team roles and document responsibilities for implementing the selected project.
Implementation Level of MLOps	PBL Activity & Project Tasks – Implementation: Evaluate different implementation levels ranging from manual ML workflows to fully automated MLOps pipelines. Student Project Task: Analyze the maturity level of the selected project and suggest improvements for automation.
Project Milestone 1	PBL Activity & Project Tasks – Implementation: Finalize the project problem statement, dataset, business objectives, architecture, team responsibilities, and implementation roadmap. Student Project Task: Submit the project proposal, dataset description, architecture diagram, and implementation plan.
Iterative Incremental Process	PBL Activity & Project Tasks – Implementation: Develop ML models through iterative improvements based on evaluation results and stakeholder feedback. Student Project Task: Maintain version-wise documentation of model improvements throughout the project.
Automation	PBL Activity & Project Tasks – Implementation: Automate data preprocessing, training, evaluation, and deployment processes using Python scripts and workflows. Student Project Task: Develop reusable automation scripts for the complete ML pipeline.
Continuous X (CI/CT/CD)	PBL Activity & Project Tasks – Implementation: Implement Continuous Integration, Continuous Training, Continuous Testing, and Continuous Deployment workflows. Student Project Task: Design and document a CI/CD workflow for the selected project.
Versioning	PBL Activity & Project Tasks – Implementation: Manage versions of datasets, source code, and trained models using Git and DVC. Student Project Task: Create version-controlled repositories and document dataset evolution.
Experiment Tracking	PBL Activity & Project Tasks – Implementation: Track experiments, hyperparameters, evaluation metrics, and artifacts using MLflow. Student Project Task: Compare multiple experiments and identify the best-performing model.
Testing	PBL Activity & Project Tasks – Implementation: Perform unit testing, data validation, integration testing, and API testing for deployed ML models.

Unit III		Student Project Task: Prepare a testing report with validation results and identified issues.
	Monitoring & ML Test Score System	PBL Activity & Project Tasks – Implementation: Monitor prediction quality, latency, model drift, and system health using monitoring dashboards. Student Project Task: Create dashboards to visualize model performance and operational metrics.
	Reproducibility & Deployment	PBL Activity & Project Tasks – Implementation: Ensure reproducible experiments and deploy trained models using Docker and FastAPI. Student Project Task: Deploy the selected model and document the deployment workflow.
	Project Milestone 2	PBL Activity & Project Tasks – Implementation: Complete version control, experiment tracking, automation pipeline, and initial deployment. Student Project Task: Submit MLflow reports, DVC pipeline, Docker implementation, and experiment comparison report.
	MLOps Development & Deployment Tools	PBL Activity & Project Tasks – Implementation: Explore Git, Docker, FastAPI, Jenkins, and GitHub Actions for building production-ready ML pipelines. Student Project Task: Integrate development and deployment tools into the project workflow.
	AutoML	PBL Activity & Project Tasks – Implementation: Automate model training and selection using AutoML frameworks and compare generated models. Student Project Task: Generate an AutoML leaderboard and analyze model performance.
	Data Validation & Data Exploration	PBL Activity & Project Tasks – Implementation: Validate data quality and perform exploratory data analysis before model training. Student Project Task: Prepare an EDA report with visualizations and validation summaries.
	Data Version Control & Data Inspection	PBL Activity & Project Tasks – Implementation: Manage dataset versions and inspect changes using DVC. Student Project Task: Create version history documentation for datasets.
	Model Serving	PBL Activity & Project Tasks – Implementation: Deploy trained models as REST APIs using FastAPI or Flask frameworks. Student Project Task: Develop and test an inference API for prediction services.
	Testing & Maintenance	PBL Activity & Project Tasks – Implementation: Perform API testing and maintain deployment logs for reliable production

Unit IV		systems.
		Student Project Task: Document maintenance activities and issue resolution reports.
	Experiment Tracking & Model Interpretability	PBL Activity & Project Tasks – Implementation: Track experiments and explain model predictions using SHAP or LIME techniques. Student Project Task: Generate interpretability reports and compare model explanations.
	Frameworks & Platforms	PBL Activity & Project Tasks – Implementation: Compare Kubeflow, MLflow, TFX, AWS SageMaker, Azure ML, and Vertex AI platforms. Student Project Task: Prepare a comparative study and recommend a suitable MLOps platform.
	Project Milestone 3	PBL Activity & Project Tasks – Implementation: Complete API deployment, experiment tracking, model registry, and explainability modules. Student Project Task: Submit API documentation, experiment reports, and platform comparison results.
	Classical Machine Learning Architecture	PBL Activity & Project Tasks – Implementation: Design a traditional machine learning architecture and identify its deployment limitations. Student Project Task: Prepare an architecture diagram and analyze workflow gaps.
	Machine Learning CV Architecture	PBL Activity & Project Tasks – Implementation: Design a Computer Vision MLOps pipeline covering data collection, training, deployment, and monitoring. Student Project Task: Build a deployment architecture for an image classification project.
	Machine Learning NLP Architecture	PBL Activity & Project Tasks – Implementation: Design an NLP MLOps architecture including preprocessing, inference, monitoring, and retraining. Student Project Task: Develop an architecture diagram for an NLP application.
	Maturity Models	PBL Activity & Project Tasks – Implementation: Study No MLOps, DevOps without MLOps, Automated Training, Automated Deployment, and Full MLOps Automated Retraining models. Student Project Task: Evaluate the maturity level of the selected project and propose improvements.
	Project Milestone 4	PBL Activity & Project Tasks – Implementation: Complete end-to-end architecture design, deployment pipeline, monitoring dashboard, and automated retraining strategy.

Unit V		Student Project Task: Submit architecture documentation and maturity assessment report.
	CI/CD Principles and Practices	PBL Activity & Project Tasks – Implementation: Design automated CI/CD pipelines for ML applications using industry best practices. Student Project Task: Create a CI/CD workflow using GitHub Actions or Jenkins.
	Integrating Machine Learning into CI/CD Pipelines	PBL Activity & Project Tasks – Implementation: Automate model training and deployment based on source code or data updates. Student Project Task: Implement an automated retraining pipeline for the project.
	Automated Testing for Machine Learning Models	PBL Activity & Project Tasks – Implementation: Perform automated unit testing, integration testing, API testing, and model validation. Student Project Task: Generate automated testing reports and validate deployment quality.
	Implementing CI/CD Workflows for Model Deployment	PBL Activity & Project Tasks – Implementation: Deploy containerized ML applications using Docker and CI/CD workflows. Student Project Task: Demonstrate automated deployment from source code commit to production inference.
	Project Milestone 5	PBL Activity & Project Tasks – Implementation: Finalize the automated MLOps pipeline including deployment, monitoring, retraining, and production validation. Student Project Task: Submit the final project report, deployment logs, CI/CD configuration files, and performance evaluation.
	Final Presentation and Viva	PBL Activity & Project Tasks – Implementation: Demonstrate the complete MLOps solution including data pipeline, experiment tracking, deployment, monitoring, and automated retraining. Student Project Task: Submit the final report, presentation slides, source code, technical documentation, and perform the live project demonstration.

Tools and Software Required: Python 3.10+, Jupyter Notebook / Google Colab / VS Code, Git & GitHub, Docker, MLflow, DVC (Data Version Control), FastAPI / Flask, GitHub Actions / Jenkins, TensorFlow / Scikit-learn, Pandas & NumPy, Matplotlib & Plotly, SHAP / LIME, Prometheus & Grafana (Monitoring), AWS SageMaker / Azure ML / Google Vertex AI (Optional Cloud Deployment)